**INTEROFFICE MEMORANDUM**

To: CISO

From: Team 02

Date: October 24, 2026

Subject: TFTP Server on Debian Splunk

## Summary:

Team 02 has successfully installed a TFTP Server within our existing infrastructure on the Debian Splunk. We have modified the firewall and security policy to allow for file upload and access to this TFTP Server's dedicated directory. This TFTP server can now be used to serve OS software updates or other files. The TFTP is only accessible from the internal network.

## Server Selected and Why

The Debian Splunk server was selected because it is in the "Internal" subnet and has sufficient storage space to hold files uploaded to the TFTP directory. The storage available on the Debian Splunk server is shown in the screenshot below:

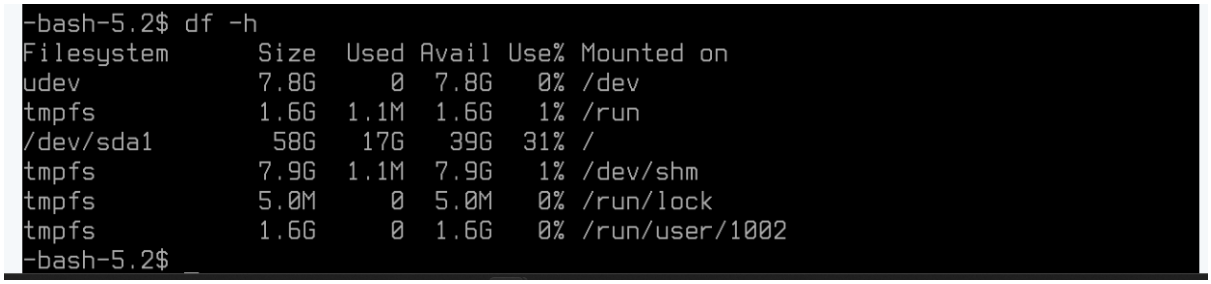



*Figure 1: Storage Capacity of Debian Splunk Server*

## Software Added

We installed the tftp-hpa package through apt install available in the Debian Linux distribution. This installs the necessary files and software to run a TFTP server. By installing this software we are able to run our TFTP server. A screenshot of the software installation is shown below:

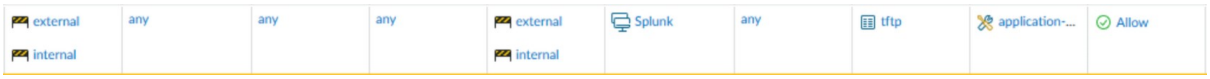
```
sudo apt update
sudo apt install tftpd-hpa
```

```
-bash-5.2$ sudo apt install tftpd-hpa
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tftpd-hpa is already the newest version (5.2+20150808-1.4).
The following packages were automatically installed and are no longer required:
  ansible-core git-man ieee-data liberror-perl patch python-babel-localedata
  python3-argcomplete python3-babel python3-distutils python3-jinja2
  python3-jmespath python3-kerberos python3-lib2to3 python3-libcloud
  python3-lockfile python3-markupsafe python3-netaddr python3-ntlm-auth
  python3-packaging python3-passlib python3-requests-kerberos
  python3-requests-ntlm python3-resolvelib python3-selinux python3-simplejson
  python3-tz python3-winrm python3-xmltodict python3-yaml
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 107 not upgraded.
-bash-5.2$
```

*Figure 2: Proof of tftp-hpa software installation*

## Firewall and Security Policy Changes

This was set through our firewall rules for all internal and external traffic. While FTP uses TCP, TFTP or Trivial File Transfer Protocol uses UDP. We allowed firewall traffic for the TFTP application in our Palo Alto Firewall.




## Security Profile of /var/lib/tftpboot

The directory used by TFTP is /var/lib/tftpboot has been set to have the least privilege needed for use which is show by the ls -la command in the screenshot below:

```
-bash-5.2$ cd /var/lib/tftpboot
-bash-5.2$ ls -la
total 16
drwxrwxrwx  2 nobody nogroup 4096 Oct 24 11:08 .
drwxr-xr-x 35 root   root    4096 Oct 24 11:00 ..
-rw-rw-rw-  1 tftp   tftp      11 Oct 24 11:09 data.txt
-rw-r--r--  1 root   root      10 Oct 24 11:08 file.txt
-bash-5.2$ _
```

*Figure 3: Security Profile of TFTP directory*

## Protections From Being Overrun With Files

To prevent our TFTP directory from being overrun with files we set up a cron job on the Debian Splunk server that is hosting the TFTP server which will run every minute and if there is a file that has been in the directory for more than 30 days it will be deleted. This will free up space so that there is enough available storage for new files.

```
0 0 * * * find {{tftp_dir}}* -type f -mtime +30 -exec rm {} \;
```

```
crontab: installing new crontab
-bash-5.2$ sudo crontab -l
*/30 * * * * /usr/bin/clamscan -r --log=/var/log/clamav/scan.log --exclude-dir="
^/sys" --exclude-dir="^/proc" --exclude-dir="^/dev" /
0 0 * * * find /var/lib/tftpboot* -type -f mtime +30 -exec rm {} \;
-bash-5.2$ _
```

*Figure 4: Cron job to free space on the Debian Splunk server*

## Evidence of the TFTP Service's Functionality

This screenshot shows our functioning TFTP service on the Debian Splunk server




*Figure 5: Connection to Debian Splunk server from the Ubuntu Workstation, file retrieved*

This screenshot shows our ability to connect to the Debian Splunk TFTP server from our Ubuntu Workstation and then retrieve a file from and upload a file to the TFTP directory




*Figure 6: File uploaded from Ubuntu Workstation on Debian Splunk server*

If there are any concerns, questions, or clarifications, please do not hesitate to reach out.

With Regards,

Team 02